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SARDAR PATEL UNIVERSITY

BCA Semester-6 External Examination – April-2025

US06CBCA53 || COMPUTER NETWORKS

Date : 03-04-2025

Time : 10.00 a.m. – 01.00 p.m.

Total : 70 Marks

Q-1 Select the appropriate option for the following questions: 10

- 1) "Interconnected collection of autonomous computers" is known as _____.
A. Computer network C. Host
B. Subnet D. Server
- 2) _____ network consists of many connection between individual parts of machines.
A. Point-to-Point network. C. People network
B. Broadcast network. D. Specialized network
- 3) _____ is a digital multiplexing technique for combining several low-rate channels into one high-rate one.
A. FDM C. TDM
B. WDM D. None
- 4) In a fiber-optic cable, the signal is propagated along the inner core by _____.
A. Refraction C. Reflection
B. Modulation D. Amplification
- 5) A _____ is a set of rules that govern data communications.
A. Layers C. Network
B. protocol D. Communication
- 6) Physical addressing and framing are responsibilities of _____.
A. Presentation layer C. Physical layer
B. Network layer D. Datalink layer
- 7) Which of the following is not a valid form of CSMA?
A. 1-persistent C. persistent
B. p-persistent D. non-persistent
- 8) A switch performs filtering on the basis of _____ address.
A. MAC C. Logical
B. Port D. Postal
- 9) A _____ cable is made of glass or plastic and transmits signals in the form of light.
A. Twisted-pair C. Coaxial cables
B. Fiber Optical D. None of these.
- 10) A list of the protocols used by a certain system, one protocol per layer, is called a _____.
A. network architecture C. Computer architecture
B. Layers stack D. protocol stack

(P.T.O.)

Q-2 Give short answers of the following questions: (Attempt TEN out of TWELVE) 20

- 1) Explain DQDB in brief.
- 2) What is computer network? What are the three categories of computer network?
- 3) List uses of computer networks.
- 4) List advantages of Fiber Optics.
- 5) Explain application of twisted pair in short.
- 6) What do you mean by Multiplexing? List out various Techniques.
- 7) Explain footprint.
- 8) Explain responsibilities of Network layer.
- 9) Explain orbit.
- 10) Write full form of CSMA. List different types of CSMA.
- 11) Explain Gateway in short.
- 12) Explain Bluetooth in short.

Q-3 A Explain LAN in detail. 5

B What are the uses/advantages & Disadvantages of computer networks? 5

OR

Q-3 A Explain WAN in detail. 5

B Explain classification of computer Network. 5

Q-4 A Explain Co-axial cable in detail. 6

B Explain Parallel transmission in detail. 4

OR

Q-4 A Explain FDM in detail. 6

B Explain Serial transmission in detail. 4

Q-5 A Explain protocol hierarchies. 5

B List all layers of OSI model. Explain any two layers. 5

OR

Q-5 A Write a note on satellite Network. 5

B Write a note on design issues for the layers. 5

Q-6 Define Topology. List various topologies. Explain Bus and Mesh topologies in detail. [10]

OR

Q-6 List Network devices. Explain Router and Bridge in detail. [10]

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